

Logic PhD Exam, September 1992

This examination is divided into 4 segments, with 3 questions in each segment. Answer any 2 questions from each segment.

1. General Logic

1. Prove that there is no single sentence ϕ in the language of ring theory such that a ring R is a model of ϕ if and only if R has characteristic 0.
2. State and prove the compactness theorem for Predicate logic. You may assume the completeness theorem.
3. State and sketch a proof of Gödel's incompleteness theorem.

2. Model Theory

1. Show that any two elementarily equivalent, countably saturated models are isomorphic.
2. State and sketch a proof of the downward Löwenheim–Skolem Theorem.
3. State the Omitting types theorem, and use it to show that any ω -complete theory of arithmetic has an ω -model.

3. Set Theory

1. Show that a countable union of countable sets is countable. Explain any use of the axiom of choice.
2. Show that $2^\kappa = \kappa^\kappa$ for all infinite κ .
3. Use Fodor's theorem to prove that the diagonal intersection of a sequence of closed, unbounded sets is closed and unbounded.

4. Recursion Theory

1. Show that a function $\phi : \omega \rightarrow \omega$ is (partial) recursive iff its graph is recursively enumerable, and that if ϕ is total and recursive then its graph is recursive.
2. Prove that there is a subset of ω which is recursively enumerable but not recursive.
3. State the normal form theorem for partial recursive functions and use it to prove the enumeration theorem.